

5.1 Accuracy of Prediction

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Prediction

- Oftentimes interest lies in **predicting** the value of a variate (the **response variate**) given the values of one or more **explanatory** variates.

- We build a **response model** that encodes how that prediction is to be carried out

$$y = \mu(\mathbf{x}) + \text{error}$$

- The explanatory variates $\mathbf{x} = (x_1, \dots, x_p)$ are used to explain or predict the values of the response.
- We use our observed data to estimate the function $\mu(\mathbf{x})$, yielding the **predictor function** $\hat{\mu}(\mathbf{x})$.

* This predictor function $\hat{\mu}(\mathbf{x})$ is then used to predict y at any given value $\mathbf{x}$.

our best guess of the relationship between y and x

- **Example:** for simple linear regression we have $\mu(x) = \alpha + \beta x$ and we might estimate the parameters of the function using least squares:

$$\hat{\mu}(x) = \hat{\alpha} + \hat{\beta}x$$

“fitted” regression line

- But how do we know if our predictions are any good? And what does “good” even mean?

Accuracy of Prediction

- Quantifying the predictive accuracy of one or more models is an extremely important task in modern-day data endeavors

– The suitability of a model may be determined entirely by its predictive accuracy.

↑ “model selection”

* So how do we measure the accuracy of a model?

– As a matter of convention, one usually measures the inaccuracy of a model's predictions.

- One measure of inaccuracy over the population $\mathcal{P}$ (of size N) is the average prediction squared error (APSE)

$$\frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \hat{\mu}(\mathbf{x}_u))^2$$

– **Note** that the quantity above is proportional to the more familiar residual sum of squares

$$\sum_{i=1}^N \hat{r}_i^2 = \sum_{i=1}^N (y_i - \hat{\mu}(\mathbf{x}_i))^2$$

which sometimes also referred to as the estimated residual mean squared error.

- This quantifies the distance between true and predicted values.

* In particular, it is the average squared distance between a response observation and its corresponding prediction.

* We might use this accuracy (inaccuracy!) measure to choose between competing models, as we will see in the sections to follow.

5.1.1 Example: Loblolly Pine Trees

- This dataset concerns the growth of Loblolly Pine Trees and contains three variates recorded for $N = 84$ trees

– **height**: the tree height in feet ✓
– **age**: the age of the tree in years ✓
– **Seed**: the seed source for the tree ✗

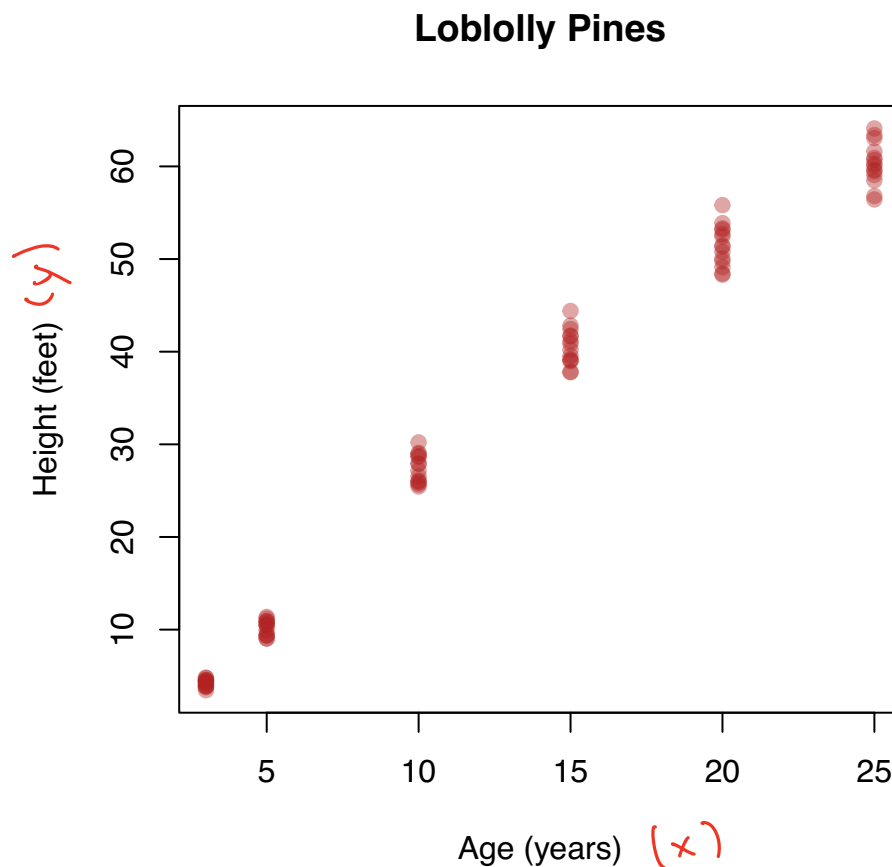
- Here is an excerpt of the data:

```
data(Loblolly)
head(Loblolly)
```

```
##      height age Seed
## 1      4.51  3  301
## 15     10.89  5  301
## 29     28.72 10  301
## 43     41.74 15  301
## 57     52.70 20  301
## 71     60.92 25  301
```

- And here is a visualization of the relationship between `height` and `age`:

```
plot(Loblolly$age, Loblolly$height, xlab="Age (years)", ylab="Height (feet)",
     main = "Loblolly Pines",
     col=adjustcolor("firebrick", 0.4), pch=19)
```



Fitting Polynomials

$$y = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3 + \dots + \alpha_p x^p + \text{error}$$

$\mu(x)$

- We might consider fitting a linear or quadratic (in age) model to this data .

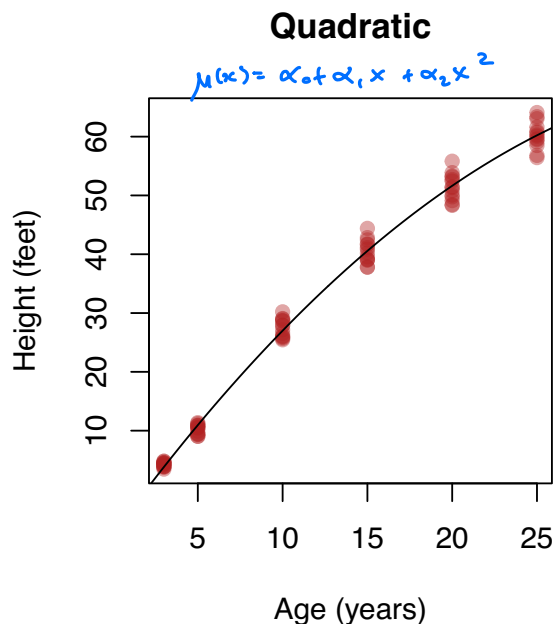
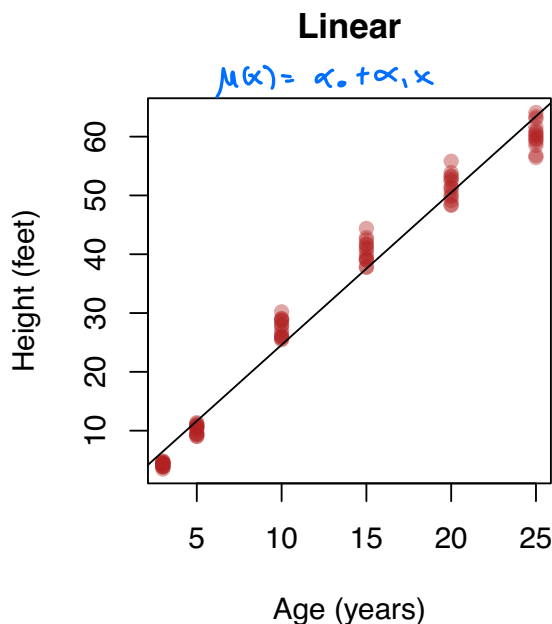
```
par(mfrow=c(1,2) )

plot(Loblolly$age[order(Loblolly$age)],
     Loblolly$height[order(Loblolly$age)],
     xlab="Age (years)", ylab="Height (feet)",
     main = "Linear",
     col=adjustcolor("firebrick", 0.4), pch=19)

lm.age = lm(height ~ age, data=Loblolly)
age.seq = seq(0, 30, by=0.1)
lines(age.seq, predict(lm.age, newdata=data.frame(age=age.seq)))

plot(Loblolly$age[order(Loblolly$age)],
     Loblolly$height[order(Loblolly$age)],
     xlab="Age (years)", ylab="Height (feet)",
     main = "Quadratic",
     col=adjustcolor("firebrick", 0.4), pch=19)

lm.age2 = lm(height ~ poly(age, 2), data=Loblolly)
lines(age.seq, predict(lm.age2, newdata=data.frame(age=age.seq)))
```



- What do you observe in these plots?

- How well do the models fit the data?

They both do a pretty good job at summarizing the relationship between height and age, but the quadratic fit appears to be slightly better because the curve intersects the points better.

- We might also consider fitting higher order polynomials as well

– we might want some additional functions to facilitate this

- `getXYpop` is a function that takes in a data frame with any number of variates and returns a data frame containing only the explanatory variate (x) and the response variate (y):

```
getXYpop <- function(xvarname, yvarname, pop) {
  popData <- pop[, c(xvarname, yvarname)]
  names(popData) <- c("x", "y")
  popData
}
```

estimates $\mu(x)$ if $\mu(x)$ is a polynomial

- `getmuhat` is a factory function that uses least squares to fit polynomial response models (for arbitrary degrees (p) in x):

```
getmuhat <- function(sampleXY, complexity) {
  formula <- paste0("y ~ ",
    if (complexity==0) "1"
    else {
      if (complexity < 20) {
        paste0("poly(x, ", complexity, ", raw = FALSE)")
        ## due to Numerical overflow
      } else {
        ## if complexity >= 20 use a spline.
        paste0("bs(x, ", complexity, ")")
      }
    }
  )

  fit <- lm(as.formula(formula), data = sampleXY)

  ## From this we construct the predictor function
  muhat <- function(x){
    if ("x" %in% names(x)) {
      ## x is a dataframe containing the variate named
      ## by xvarname
      newdata <- x
    } else
      ## x is a vector of values that needs to be a data.frame
    {newdata <- data.frame(x = x) }
```

```

## The suppress warnings prediction
suppressWarnings({
  ypred = predict(fit, newdata = newdata, silent = TRUE) })

  ypred
}
## muhat is the function that we need to calculate values
## at any x, so we return this function from getmuhat
muhat
}

```

- plotLoblollyfit is a function that plots the Loblolly Pine data and overlays the fitted model muhat

```

plotLoblollyfit <- function(muhat, complexity=NULL) {

  if ( is.null(complexity) ) title = bquote(hat(mu) ~ "(Piecewise)")
  else title = bquote(hat(mu) ~ "(degree =" ~ .(complexity) * ")")

  plot(Loblolly[,c("age", "height")],
       main= title,
       xlab = "Age (years)", ylab = "Height (feet)",
       pch=19, col= adjustcolor("black", 0.5))

  xlim = extendrange(Loblolly[, "age"])
  curve(muhat, from = xlim[1], to = xlim[2],
        add = TRUE, col="steelblue", lwd=2, n=1000)

}

```

;

- Let's use these functions to fit and plot the quadratic model:

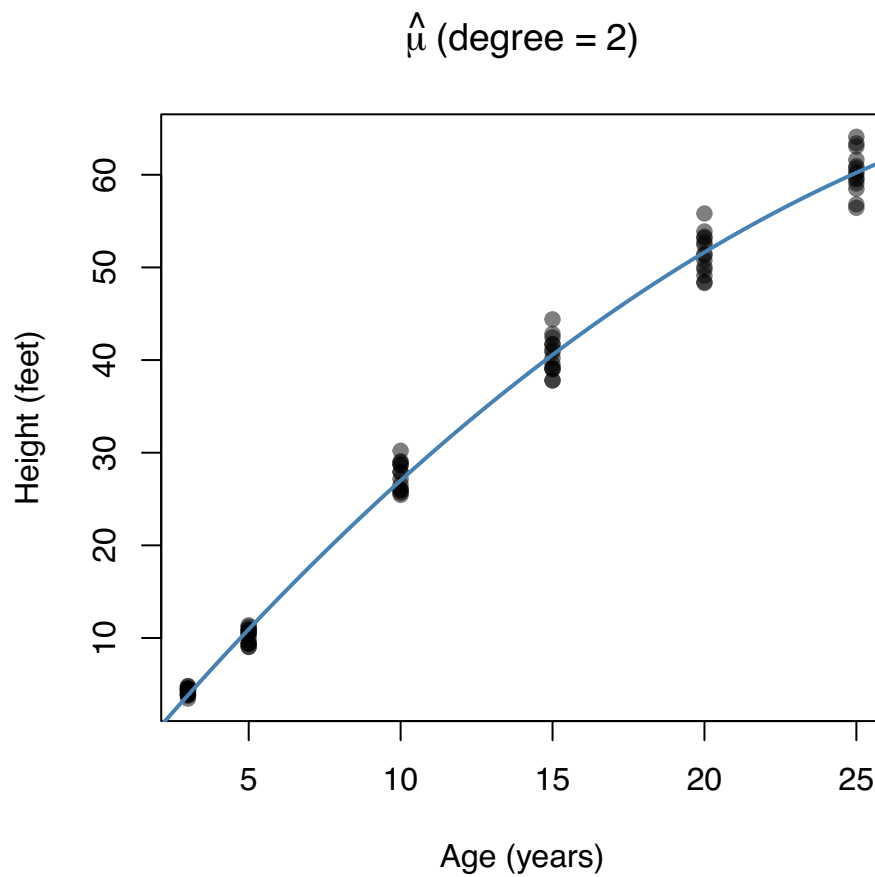
```

Lobpop <- getXYpop(xvarname="age",
                  yvarname="height",
                  pop=Loblolly)

muhat <- getmuhat(Lobpop, complexity=2)

plotLoblollyfit(muhat, complexity=2)

```



- Let's now explore fitting higher order polynomials

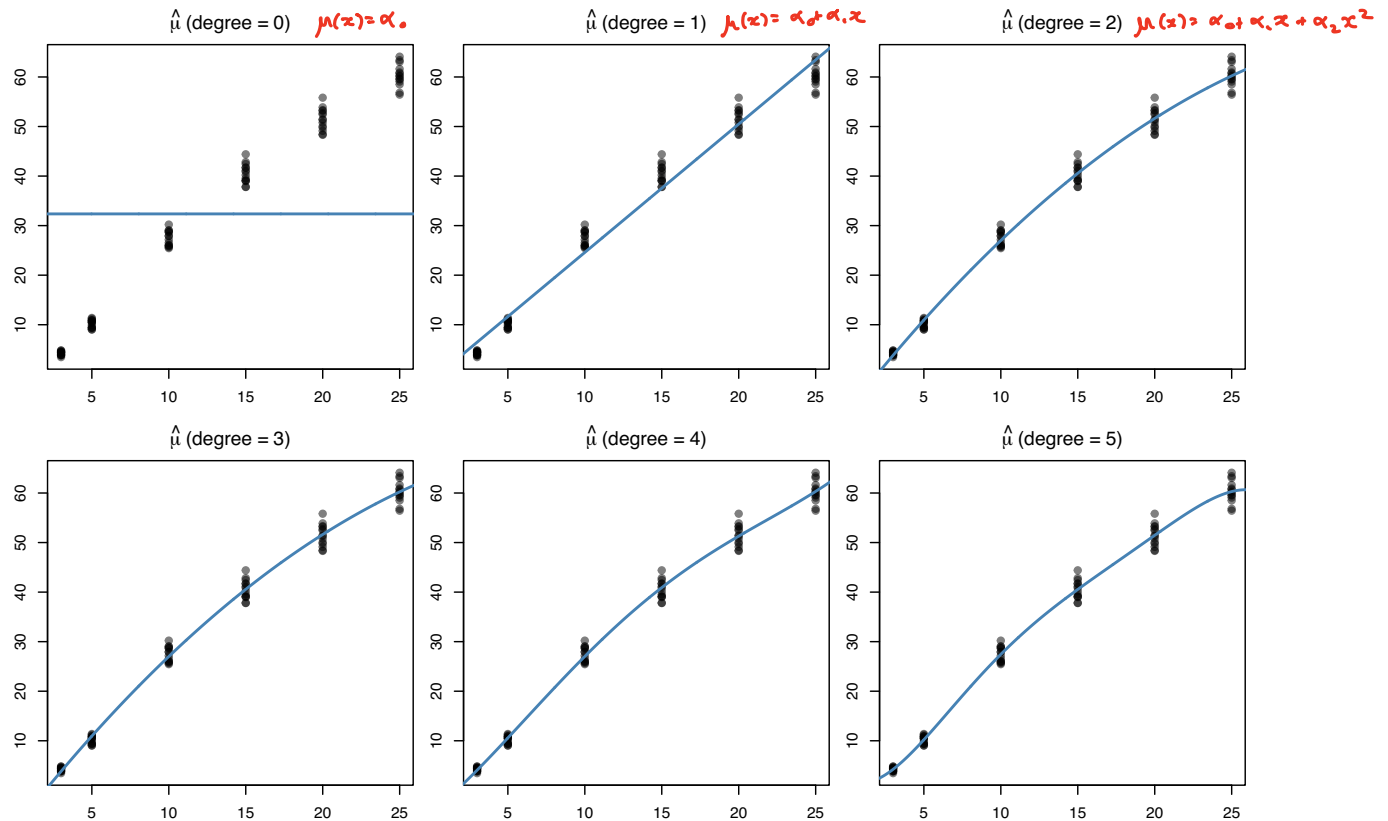
```
par(mfrow=c(2,3), mar=2.5*c(1,1,1,0.1))
```

```
dset = 0:5
```

p = 0, 1, 2, 3, 4, 5

```
muhats = lapply(dset, getmuhat, sampleXY=Lobpop )
```

```
for (deg in dset) plotLobollyfit(muhats[[deg+1]], complexity = deg)
```



- What do you observe in these plots?
- How well do the models fit the data?

All models with degree ≥ 2 fit the data well, but higher degrees don't seem to fit that much better.

- We can use the APSE

$$\frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \hat{\mu}(x_u))^2$$

to quantify the accuracy of these models

- The function **apse** will perform this calculation for us.

```

apse <- function(y, x, predfun) {
  {mean((y - predfun(x))^2, na.rm = TRUE)}
}

```

$\hat{\mu}(x)$

- The APSE calculated over the entire population (for each model) is shown below:


```
dset = 0:5
Lob.apse = unlist( lapply(muhats, apse,
                          y= Lobpop$y, x =Lobpop$x ))

temp= rbind( 0:5, round(unlist(Lob.apse), 2) )
row.names(temp) = c("Polynomial Degree", "APSE")
dimnames(temp)[[2]] = rep("", 6)
temp
```

##		<u>Increasing Model Complexity</u>					
## Polynomial Degree	0.00	1.00	2.00	3.00	4.00	5.00	
## APSE	422.31	8.48	2.79	2.79	2.72	2.64	

Increasing Prediction Accuracy →

- Which model is best?
 - How much do we gain by making the model more complex (i.e., increasing degree of the polynomial)?
- Strictly speaking Degree 5 has the smallest APSE, but it's not that much smaller than the APSE with degree 2. I would choose to use the quadratic model.

Fitting Piecewise Functions

- To model the relationship between height and age we could also fit a piecewise constant function
 - where different x intervals define different “pieces”
 - and the predictor function is based on the average y value in each “piece”
- Such an approach is especially relevant here because $x = \text{age}$ only has 6 unique values:

```
unique(Loblolly$age)
```

```
## [1] 3 5 10 15 20 25
```

- We will consider three approaches to this:
 - piecewise constant with linear interpolation (`getmuFun`)
 - piecewise constant with each unique age defining the left endpoint of a “piece” (`getmuFun2`)
 - piecewise constant with each unique age in the middle of a “piece” (`getmuFun3`)
- Here we define `getmuFun`

```
getmuFun <- function(pop, xvarname, yvarname){
  pop = na.omit(pop[, c(xvarname, yvarname)])

  # rule = 2 means return the nearest y-value when extrapolating, same as above.
  # ties = mean means that repeated x-values have their y-values averaged, as above.
  muFun = approxfun(pop[,xvarname], pop[,yvarname], rule = 2, ties = mean)
  return(muFun)
}
```

- Here we define getmuFun2

```
getmuFun2 <- function(pop, xvarname, yvarname){
  pop = na.omit(pop[, c(xvarname, yvarname)])

  # rule = 2 means return the nearest y-value when extrapolating, same as above.
  # ties = mean means that repeated x-values have their y-values averaged, as above.
  muFun = approxfun(pop[,xvarname], pop[,yvarname], method = "constant", rule = 2, ties = mean, f=1/2)
  return(muFun)
}
```

- Here we define getmuFun3

```
getmuFun3 <- function(pop, xvarname, yvarname){
  ## First remove NAs
  pop <- na.omit(pop[, c(xvarname, yvarname)])
  x <- pop[, xvarname]
  y <- pop[, yvarname]
  xks <- unique(x)
  muVals <- sapply(xks,
    FUN = function(xk) {
      mean(y[x==xk])
    })
  ## Put the values in the order of xks
  ord <- order(xks)
  xks <- xks[ord]
  xkRange <- xks[c(1,length(xks))]
  minxk <- min(xkRange)
  maxxk <- max(xkRange)
  ## mu values
  muVals <- muVals[ord]
  muRange <- muVals[c(1, length(muVals))]
  muFun <- function(xVals){
    ## vector of predictions
    ## same size as xVals and NA in same locations
    predictions <- xVals
    ## Take care of NAs
    xValsLocs <- !is.na(xVals)
    ## Just predict non-NA xVals
```

```

predictions[xValsLocs] <- sapply(xVals[xValsLocs],
  FUN = function(xVal) {
    if (xVal < minxk) {
      result <- muRange[1]
    } else if (xVal > maxxk) {
      result <- muRange[2]
    } else if ( any(xVal == xks) ) {
      result <- muVals[xks == xVal]
    } else {
      xlower <- max(c(minxk, xks[xks < xVal]))
      xhigher <- min(c(maxxk, xks[xks >= xVal]))
      mulower <- muVals[xks == xlower]
      muhigher <- muVals[xks == xhigher]

      midx = (xlower + xhigher)/2
      if (xVal <= midx) result <- mulower
      else result <- muhigher
    }
    result
  }
)
## Now return the predictions (including NAs)
predictions
}
muFun
}

```

- The three functions fitted to the population are visualized below

```

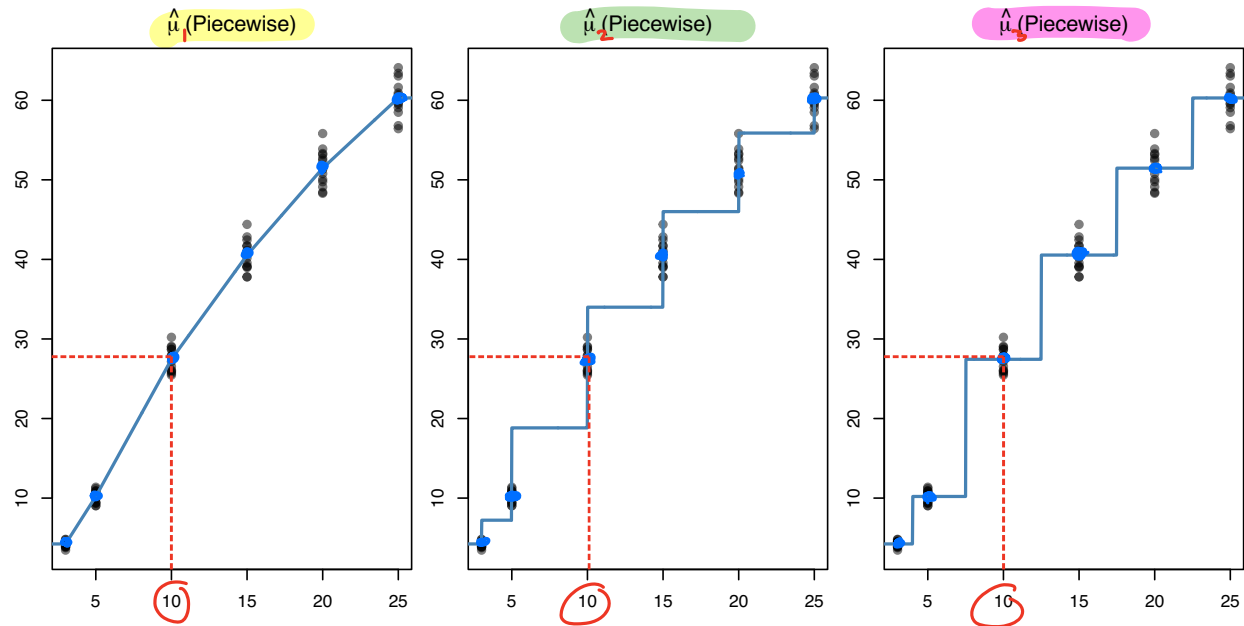
par(mfrow=c(1,3), mar=2.5*c(1,1,1,0.1))

mu.age = getmuFun(pop=Loblolly,
                  xvarname="age", yvarname="height")
plotLoblollyfit(mu.age)

mu.age2 = getmuFun2(pop=Loblolly,
                   xvarname="age", yvarname="height")
plotLoblollyfit(mu.age2)

mu.age3 = getmuFun3(pop=Loblolly,
                   xvarname="age", yvarname="height")
plotLoblollyfit(mu.age3)

```



* To better understand how these three functions differ, let's evaluate them just before, at and after $x = 10$:

```
x = c(9.9, 10, 10.1)
round(cbind(x, mu=mu.age(x), mu2=mu.age2(x), mu3=mu.age3(x)), 2)
```

```
##      x      mu      mu2      mu3
## [1,] 9.9 27.10 18.82 27.44
## [2,] 10.0 27.44 27.44 27.44
## [3,] 10.1 27.70 33.99 27.44
```

- Which function fits the population better?

For the observed data these functions are equivalent. But they differ with respect to the values of x not observed. Which is best depends on the relationship between y and x you expect.

5.1.2 Example: Global Temperature Data

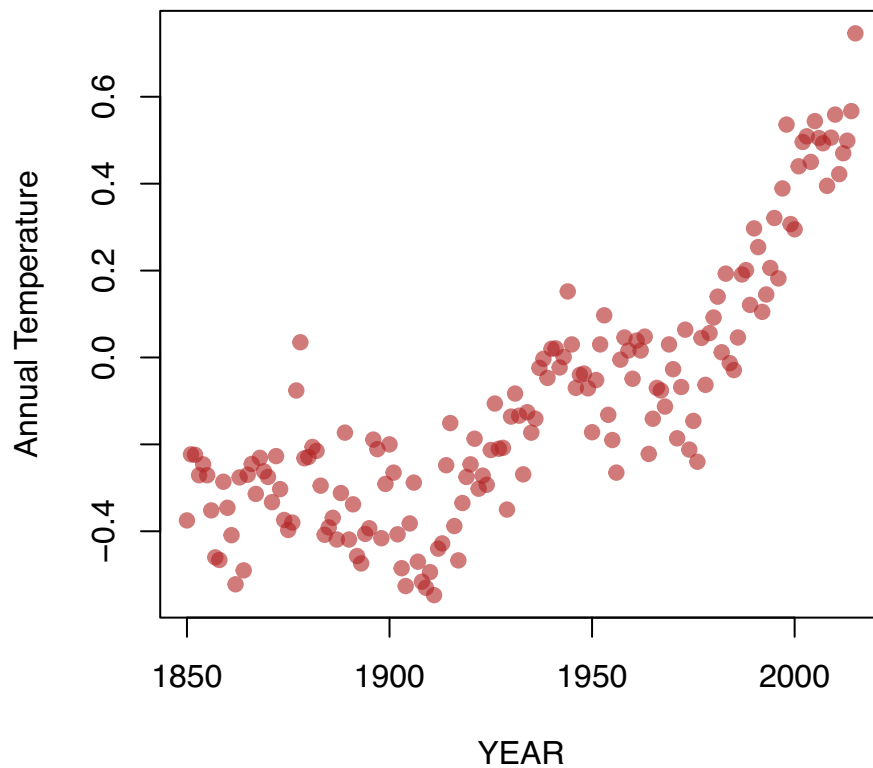
- This dataset concerns annual temperatures of the planet over time. The variables of interest are $y =$ ANNUAL temperature and $x =$ YEAR
 - Let's load and visualize it:

```
temperature <- read.csv("/Users/nstevens/Dropbox/Teaching/STAT_341/Fall_2023/Lectures/Data/temperature.csv")
head(temperature)
```

```
##  YEAR      JAN      FEB      MAR      APR      MAY      JUN      JUL      AUG      SEP      OCT
## 1 1850 -0.702 -0.284 -0.732 -0.570 -0.325 -0.213 -0.128 -0.233 -0.444 -0.452
## 2 1851 -0.303 -0.362 -0.485 -0.445 -0.302 -0.189 -0.215 -0.153 -0.108 -0.063
## 3 1852 -0.308 -0.477 -0.505 -0.559 -0.209 -0.038 -0.016 -0.195 -0.125 -0.216
## 4 1853 -0.177 -0.330 -0.318 -0.352 -0.268 -0.179 -0.059 -0.148 -0.409 -0.359
## 5 1854 -0.360 -0.280 -0.284 -0.349 -0.230 -0.215 -0.228 -0.163 -0.115 -0.188
## 6 1855 -0.176 -0.400 -0.303 -0.217 -0.336 -0.160 -0.268 -0.159 -0.339 -0.211
##      NOV      DEC ANNUAL
## 1 -0.190 -0.268 -0.375
## 2 -0.030 -0.067 -0.223
## 3 -0.187  0.083 -0.224
## 4 -0.256 -0.444 -0.271
## 5 -0.369 -0.232 -0.246
## 6 -0.212 -0.510 -0.271
```

```
plot(temperature$YEAR, temperature$ANNUAL, xlab="YEAR", ylab="Annual Temperature",
     main="Global Annual Temperature", col=adjustcolor("firebrick", 0.6), pch=19 )
```

Global Annual Temperature



- In what follows we shall fit a variety of polynomials and piecewise functions to this data and use APSE to determine which model(s) seem appropriate

Fitting Polynomials

- To begin with, let's define `plotTemperaturefit`, a function that plots the Temperature data and overlays the fitted model `muhat`

```
plotTemperaturefit <- function(muhat, complexity=NULL) {

  if ( is.null(complexity) ) title = ""
  else title = paste0("polynomial degree=", complexity,"")

  plot(temperature[,c("YEAR", "ANNUAL")],
       main= title,
       xlab = "age", ylab = "height",
       pch=19, col= adjustcolor("Grey", 0.8))

  xlim = extendrange(temperature[, "YEAR"])
  curve(muhat, from = xlim[1], to = xlim[2],
        add = TRUE, col="steelblue", lwd=2, n=1000)

}
```

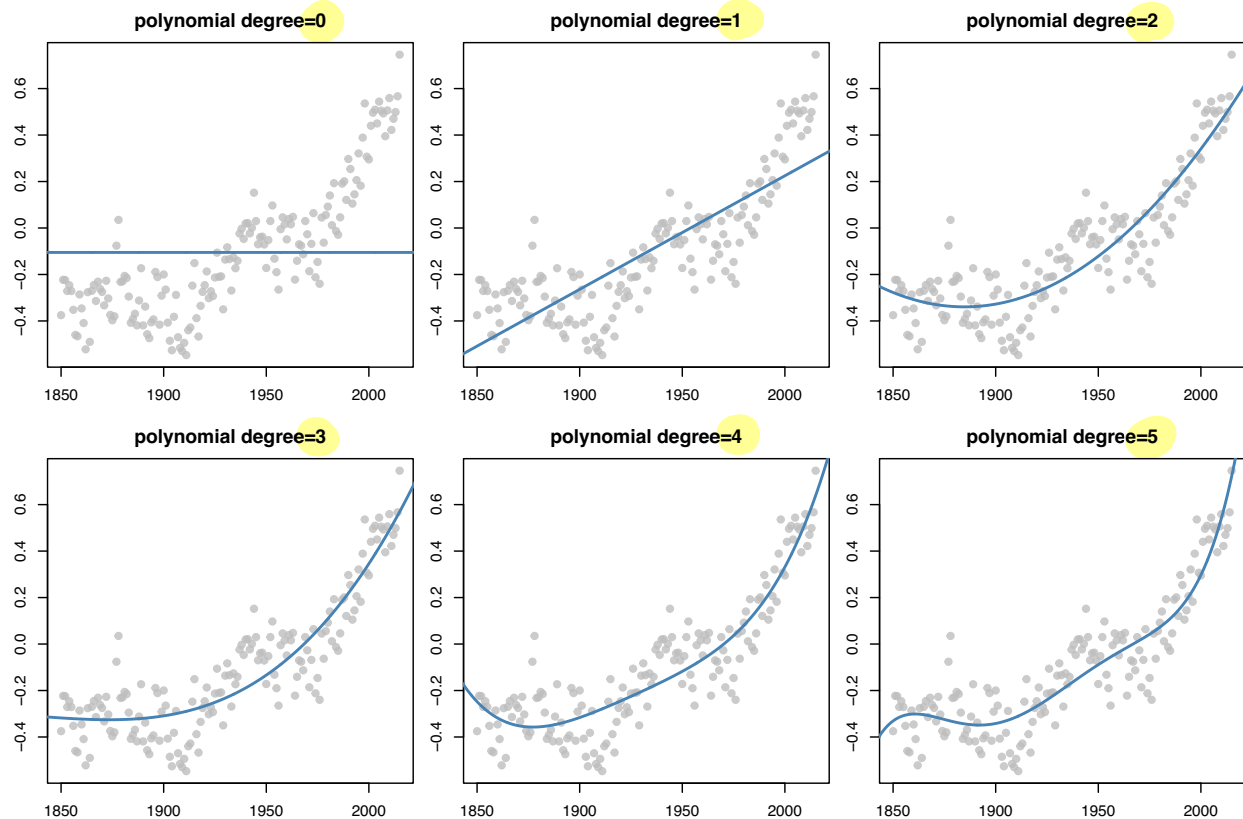
Polynomial Regression (low degree)

- Consider fitting polynomials of degree 0:5:

```
par(mfrow=c(2,3), mar=2.5*c(1,1,1,0.1))

temppop= getXYPop(xvarname="YEAR", yvarname="ANNUAL",pop=temperature)
dset = 0:5
muhats = lapply(dset, getmuhat, sampleXY=temppop)

for (i in 1:length(dset) ) plotTemperaturefit(muhats[[i]], dset[i])
```



- The APSE calculated over the entire population (for each model) is shown below:

```
dset = 0:5
temp.apse = unlist( lapply(muhats, apse,
                           y= temppop$y, x =temppop$x ))

tab = rbind( 0:5, round(unlist(temp.apse), 4) )
row.names(tab) = c("Polynomial Degree", "APSE")
dimnames(tab)[[2]] = rep("", 6)
tab
```

	0	1	2	3	4	5
Polynomial Degree	0.0000	1.0000	2.0000	3.0000	4.0000	5.0000
APSE	0.0807	0.0258	0.0148	0.0145	0.0139	0.0132

Increasing Model Complexity →

← Increasing Predictive Accuracy

- Which model is best? Degree 5
- How much do we gain by making the model more complex (i.e., increasing degree of the polynomial)?

A little bit

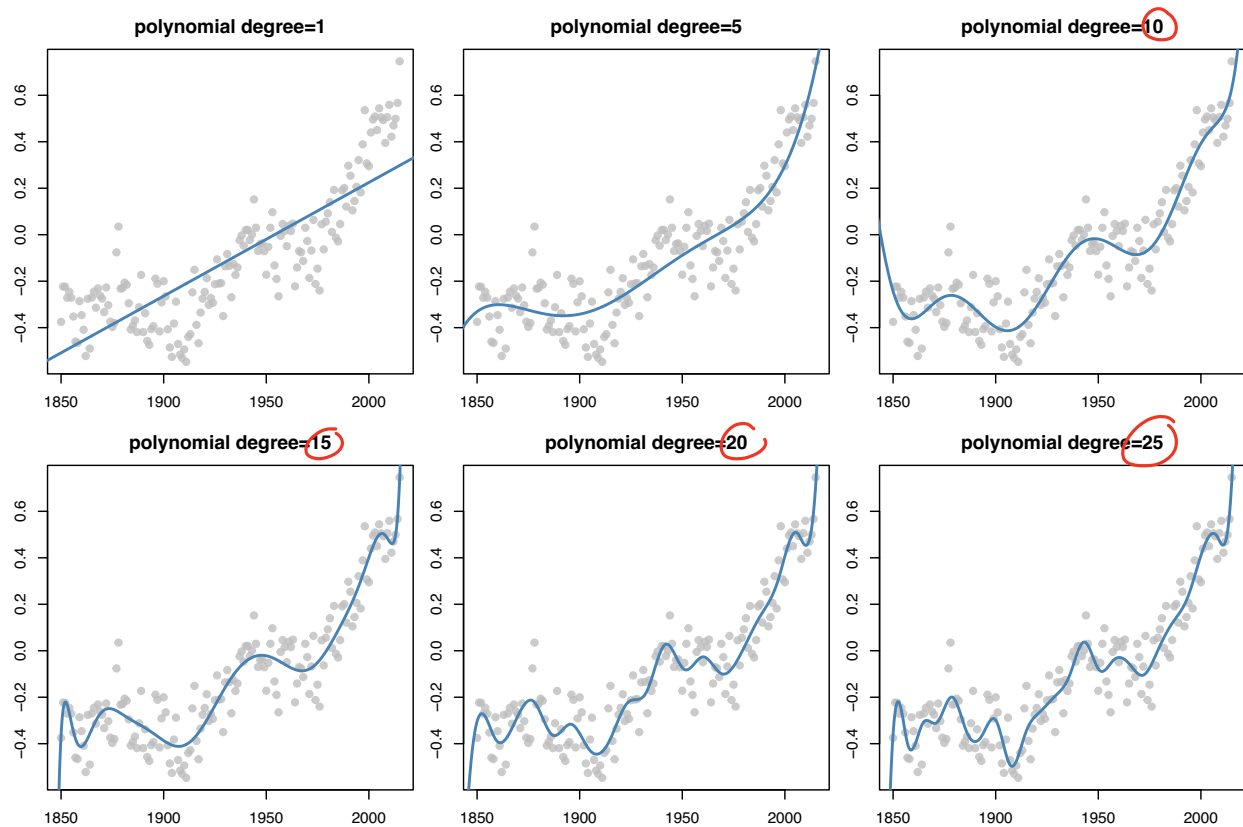
Polynomial Regression (high degree)

- Consider fitting polynomials of degrees 1, 5, 10, 15, 20 and 25:

```
par(mfrow=c(2,3), mar=2.5*c(1,1,1,0.1))

dset = c(1, 5, 10, 15, 20, 25)

muhats = lapply(dset, getmuhat, sampleXY=temppop )
for (i in 1:length(dset) ) plotTemperaturefit(muhats[[i]], dset[i])
```



- The APSE calculated over the entire population (for each model) is shown below:

```
temp.apse = unlist( lapply(muhats, apse,
                           y= temppop$y, x =temppop$x ))

dset = c(1, 5, 10, 15, 20, 25)

tab = rbind( dset, round(unlist(temp.apse), 4) )
row.names(tab) = c("Polynomial Degree", "APSE")
dimnames(tab)[[2]] = rep("", 6)
tab
```


	Increasing Complexity →					
## Polynomial Degree	1.0000	5.0000	1e+01	15.0000	20.0000	25.0000
## APSE	0.0258	0.0132	9e-03	0.0084	0.0077	0.0071
	Decreasing APSE →					

- Which model is best? *Degree 25*
- How much do we gain by making the model more complex (i.e., increasing degree of the polynomial)?
Not much

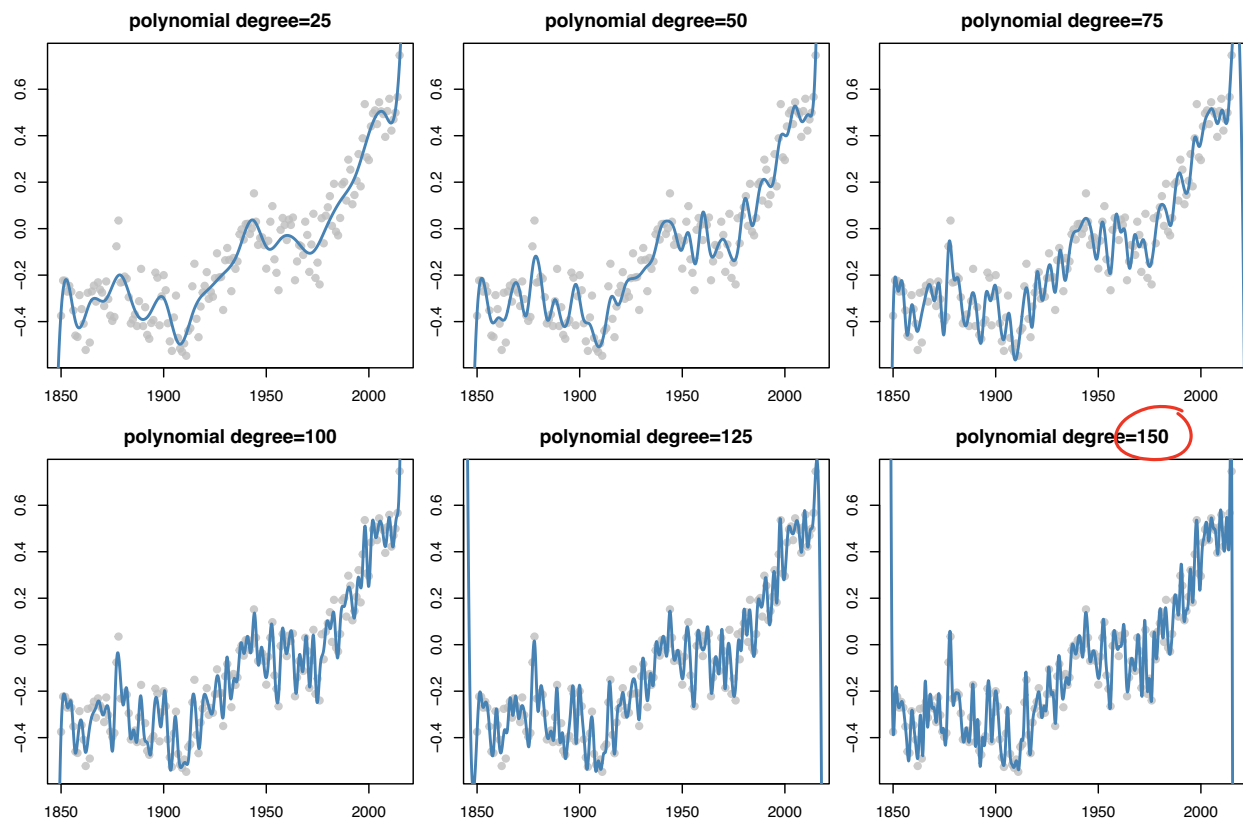
Polynomial Regression (higher degree)

- Consider fitting a polynomials of degree 20, 50, 75, 100, 125 and 150 to Global Annual Temperature.

```
par(mfrow=c(2,3), mar=2.5*c(1,1,0.1))

dset = c(25, 50, 75, 100, 125, 150)
muhat = lapply(dset, getmuhat, sampleXY=temppop )

for (i in 1:length(dset)) plotTemperaturefit(muhat[[i]], dset[i])
```



- The APSE calculated over the entire population (for each model) is shown below:

```
dset = c(25, 50, 75, 100, 125, 150)
temp.apse = unlist( lapply(mu_hats, apse,
                           y= temppop$y, x =temppop$x ))

tab = rbind( dset, round(unlist(temp.apse), 5) )
row.names(tab) = c("Polynomial Degree", "APSE")
dimnames(tab)[[2]] = rep("", 6)
tab
```

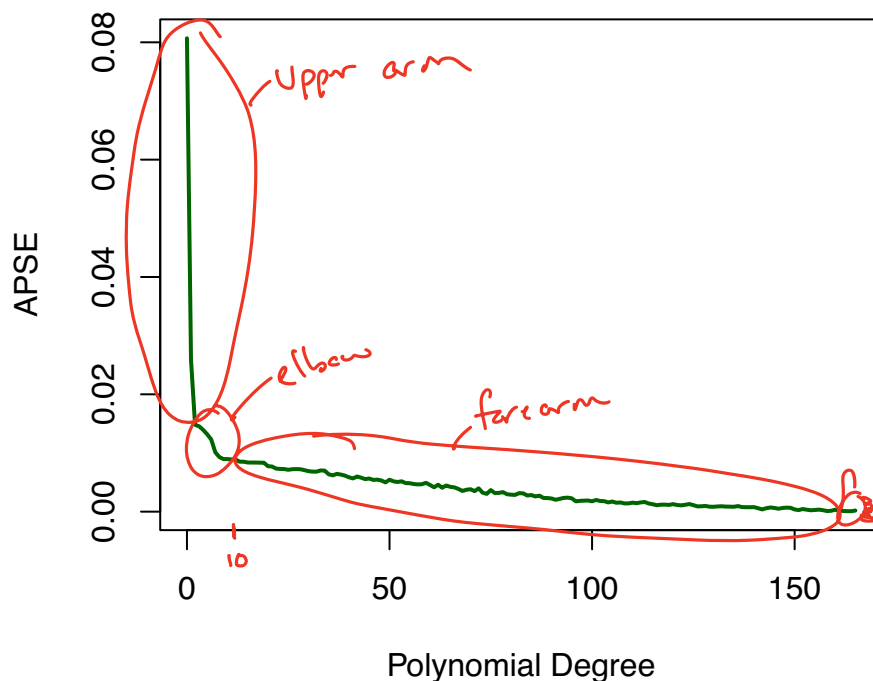
```
##
## Polynomial Degree 25.00000 50.00000 75.00000 1.00e+02 1.25e+02 1.5e+02
## APSE              0.00714  0.00544  0.00368  1.76e-03 1.11e-03 3.7e-04
```

Degree = 150

- Which model is best? Degree 150
- How much do we gain by making the model more complex (i.e., increasing degree of the polynomial)?
- What have we learned?

In general, as we increase model complexity (i.e., degree) the predictor function becomes highly tuned to every subtle variation in the data.

- So what level of *complexity* (i.e. what degree of polynomial) is appropriate?
 - To answer this question, let's visualize the improvement in APSE achieved by iteratively increasing the polynomial's degree:



- What APSE value is acceptable?

- The plot above indicates minimal improvement in APSE by choosing degrees > 10 .

- We should choose a complexity near the elbow to balance parsimony and accuracy.

Fitting Piecewise Functions

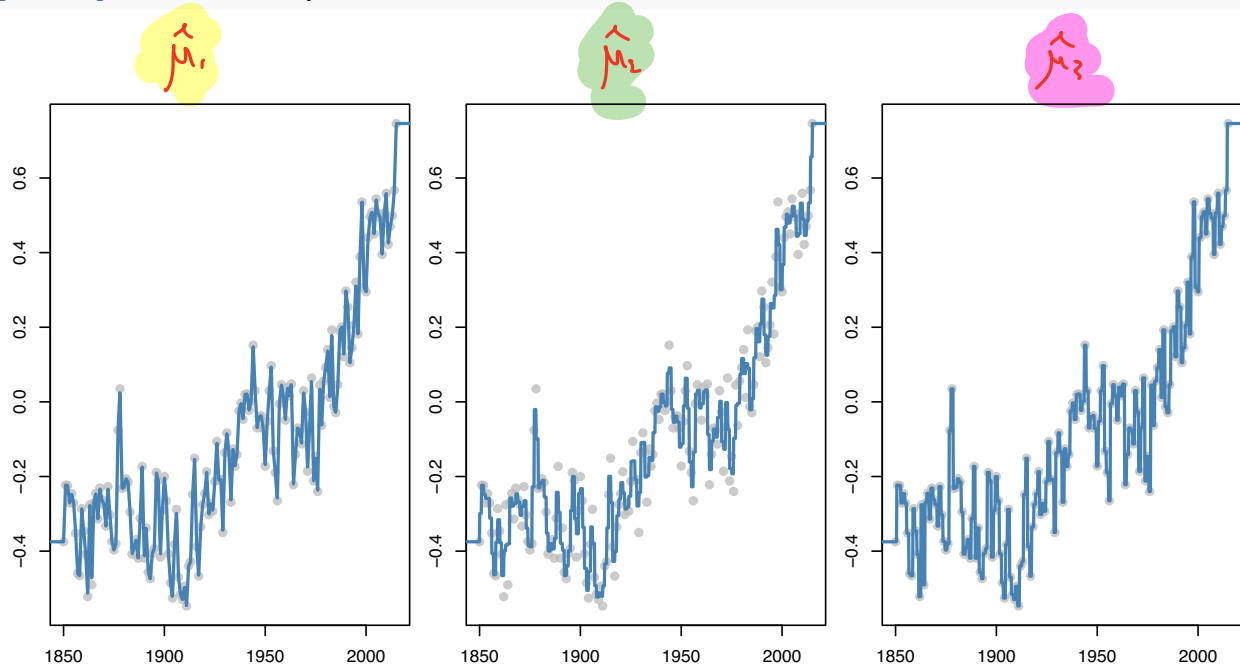
- To model this relationship we could also fit the piecewise constant functions:

```
par(mfrow=c(1,3), mar=2.5*c(1,1,1,0.1))

mu.year = getmuFun(pop=temperature,
                  xvarname="YEAR", yvarname="ANNUAL")
plotTemperaturefit(mu.year)

mu.year2 = getmuFun2(pop=temperature,
                    xvarname="YEAR", yvarname="ANNUAL")
plotTemperaturefit(mu.year2)

mu.year3 = getmuFun3(pop=temperature,
                    xvarname="YEAR", yvarname="ANNUAL")
plotTemperaturefit(mu.year3)
```



- * Comparing these three plots, what do you observe? ← they're all very similar
- * Is this level of *complexity* appropriate? ← no, these models seem more complicated than necessary.

5.1.3 Measuring Inaccuracy (Fairly)

- So far, we have been fitting models of various complexities to data, and comparing them on the basis of their **average prediction squared error (APSE)**:

$$\frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \hat{\mu}(\mathbf{x}_u))^2$$

- HOWEVER:** In this approach we estimate the predictor function and measure its accuracy using the **exact same set** of observations.

- Our current inaccuracy measure can thus be written as

$$APSE(\mathcal{P}, \hat{\mu}_{\mathcal{P}}) = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \hat{\mu}(\mathbf{x}_u))^2 = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \hat{\mu}_{\mathcal{P}}(\mathbf{x}_u))^2$$

where the notation $\hat{\mu}_{\mathcal{P}}(\mathbf{x}_u)$ emphasizes the fact that the predictor function was determined from the entire population.

- This isn't the most honest way to estimate a predictor function's accuracy.
 - This method will **underestimate** the APSE for **predictions** at values of x not existing in the data (i.e., new/different values).
 - * The dataset used to fit (**train**) the model is the same dataset we use to evaluate (**test**) the model.
 - * This (bad) approach leads to a problem known as **overfitting** which will be discussed in more detail below.

This is the phenomenon where we finely tune a model to a specific dataset whereby the model performs well on this data but poorly on other data

- Ideally (to provide a fair evaluation of prediction accuracy) we would use different data to *train* vs. *test* the model and our measure of inaccuracy would reflect this.
 - We should estimate the predictor function using a sample $\mathcal{S}$ (sometimes called the **training set**)
 - AND measure the inaccuracy over the population $\mathcal{P}$, or over the units in the population not included in the sample: $\mathcal{T} = \mathcal{P} \setminus \mathcal{S}$ (sometimes called the **test set**)

- In this case we could write the APSE as

$$APSE(\mathcal{P}, \hat{\mu}_{\mathcal{S}}) = \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \hat{\mu}_{\mathcal{S}}(\mathbf{x}_u))^2$$

* *

– This notation emphasizes that the estimate of the predictor function $\hat{\mu}$ is based on a sample $\mathcal{S}$

- Since $\mathcal{P} = \mathcal{S} \cup \mathcal{T}$ and $\mathcal{S} \cap \mathcal{T} = \emptyset$ (i.e., $\mathcal{T}$ is the complement set of $\mathcal{S}$ in $\mathcal{P}$) the APSE as defined in this way can be decomposed into a sum of two pieces:

$$\begin{aligned} APSE(\mathcal{P}, \hat{\mu}_{\mathcal{S}}) &= \frac{1}{N} \sum_{u \in \mathcal{P}} (y_u - \hat{\mu}_{\mathcal{S}}(\mathbf{x}_u))^2 \\ &= \left(\frac{n}{N}\right) \frac{1}{n} \sum_{u \in \mathcal{S}} (y_u - \hat{\mu}_{\mathcal{S}}(\mathbf{x}_u))^2 + \left(\frac{N-n}{N}\right) \frac{1}{N-n} \sum_{u \in \mathcal{T}} (y_u - \hat{\mu}_{\mathcal{S}}(\mathbf{x}_u))^2 \\ &= \underbrace{\left(\frac{n}{N}\right) APSE(\mathcal{S}, \hat{\mu}_{\mathcal{S}})} + \underbrace{\left(\frac{N-n}{N}\right) APSE(\mathcal{T}, \hat{\mu}_{\mathcal{S}})} \end{aligned}$$

- Given that interest often lies in the quality of the predictions **outside of the sample**

– we might exclusively calculate average prediction squared error over $\mathcal{T}$

$$APSE(\mathcal{T}, \hat{\mu}_{\mathcal{S}}) = \frac{1}{N-n} \sum_{u \in \mathcal{T}} (y_u - \hat{\mu}_{\mathcal{S}}(\mathbf{x}_u))^2. \quad *$$

* But clearly if $n \ll N$, the value $APSE(\mathcal{T}, \hat{\mu}_{\mathcal{S}})$ will not be that different from $APSE(\mathcal{P}, \hat{\mu}_{\mathcal{S}})$.

Example: Global Temperature Data

- To begin with we will define some functions that will be useful throughout the example
- `getSampleComp` is a function that given a population (`pop`) of size N and a sample size n will return an N element vector containing n TRUE and $N - n$ FALSE values
 - TRUE indicates the inclusion of a unit in the sample
 - FALSE indicates the inclusion of a unit in the test set

```
getSampleComp <- function(pop, size, replace=FALSE) {
  N <- popSize(pop)
  samp <- rep(FALSE, N)
  samp[sample(1:N, size, replace = replace)] <- TRUE
  samp
}
```

- `getXYSample` is a function that extracts a given sample of units from a given population and returns a data frame containing just the explanatory variate (x) and the response variate (y)

```
getXYSample <- function(xvarname, yvarname, samp, pop) {
  sampData <- pop[samp, c(xvarname, yvarname)]
  names(sampData) <- c("x", "y")
  sampData
}
```

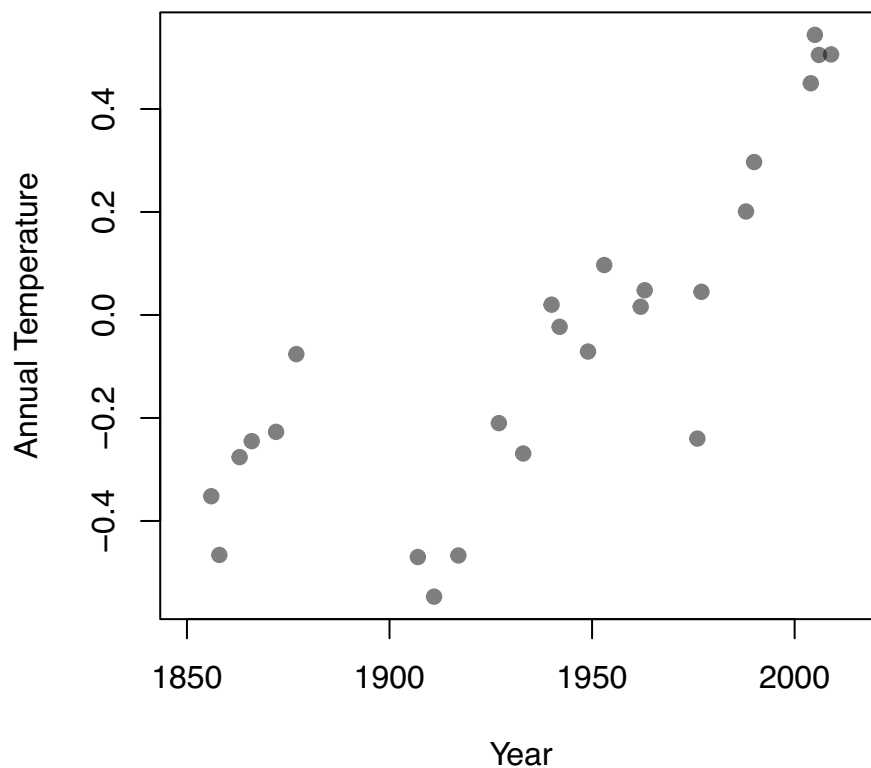
- popSize and sampSize are functions used by both functions above to calculate the population and sample sizes, respectively.

```
popSize <- function(pop) {nrow(as.data.frame(pop))}
sampSize <- function(samp) {popSize(samp)}
```

- Suppose we have the following sample of $n = 25$ observations.

```
set.seed(341)
sample.temperature <- getSampleComp(temperature, 25)
sample.Data = getXYSample("YEAR", "ANNUAL",
                           samp=sample.temperature, pop=temperature)
plot(sample.Data,
     main= "Sample Data",
     xlab = "Year", ylab = "Annual Temperature",
     pch=19, col= adjustcolor("black", 0.5), xlim=range(temperature$YEAR))
```

Sample Data (5) $n = 25$



- Let's try fitting some models to this sample data, and evaluate them on both the sample and the test data
- To do this, the following function will be useful
 - `plotTemperatureSTfit` fits a model to $\mathcal{S}$ and plots both $\mathcal{S}$ and $\mathcal{T}$ with the fitted model overlaid on each

```
plotTemperatureSTfit <- function(muhat=NULL, Sam=NULL, complexity=NULL) {

  if ( is.null(complexity) ){
    titleS <- bquote(hat(mu) ~ "(piecewise) on S")
    titleT <- bquote(hat(mu) ~ "(piecewise) on T")
  }else{
    titleS <- bquote(hat(mu) ~ "(degree =" ~ .(complexity) * ")" ~ "on S")
    titleT <- bquote(hat(mu) ~ "(degree =" ~ .(complexity) * ")" ~ "on T")
  }
  title = c(paste0("muhat (degree=", complexity,") on S"),
            paste0("muhat (degree=", complexity,") on T"))

  xlim <- extendrange(temperature[, "YEAR"])
  ylim <- extendrange(temperature[, "ANNUAL"])

  plot(temperature[Sam,c("YEAR", "ANNUAL")],
       main= titleS,
       xlab = "Year", ylab = "Annual Temperature",
       pch=19, col= adjustcolor("black", 0.5),
       xlim=xlim, ylim=ylim)

  curve(muhat, from = xlim[1], to = xlim[2], add = TRUE,
        col="steelblue", lwd=2, n=1000)

  Tpop = !Sam

  plot(temperature[Tpop, c("YEAR", "ANNUAL")],
       main=titleT,
       xlab = "Year", ylab = "Annual Temperature",
       pch=19, col= adjustcolor("black", 0.5),
       xlim=xlim, ylim=ylim)

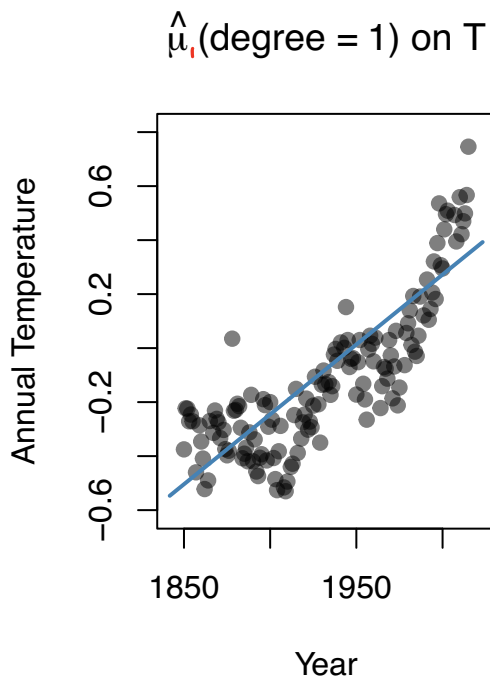
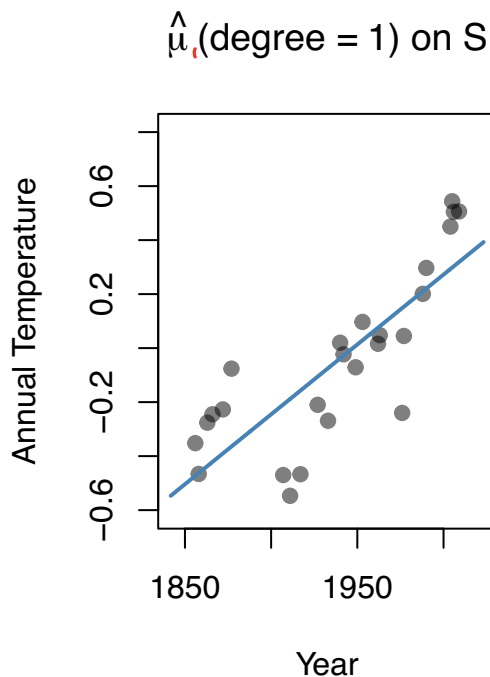
  curve(muhat, from = xlim[1], to = xlim[2], add = TRUE,
        col="steelblue", lwd=2, n=1000)

}
```

- Here is a linear predictor function $\mu(x) = \alpha_0 + \alpha_1 x$ (degree = 1)
 - Recall: the line is fitted based on $\mathcal{S}$, and plotted on both $\mathcal{S}$ and $\mathcal{T}$

```
muhat <- getmuhat(sample.Data, 1)

par(mfrow=c(1,2))
plotTemperatureSTfit(muhat, sample.temperature, 1)
```



- The APSE calculated over the sample $\mathcal{S}$ is:

```
apse(y = temperature[sample.temperature, "ANNUAL"],
     x = temperature[sample.temperature, "YEAR"], predfun = muhat)
```

```
## [1] 0.03482818
```

← APSE($\mathcal{S}$, $\hat{\mu}_\mathcal{S}$)

- The APSE calculated over the complement set $\mathcal{T} = \mathcal{P} \setminus \mathcal{S}$ is:

```
Tpop <- !sample.temperature
apse(y = temperature[Tpop, "ANNUAL"], x = temperature[Tpop, "YEAR"], predfun = muhat)
```

```
## [1] 0.02540705
```

← APSE($\mathcal{T}$, $\hat{\mu}_\mathcal{S}$)

- Note that the APSE over $\mathcal{S}$ is ~~less~~ higher than the APSE over $\mathcal{T}$ (which is in fact not to be expected)

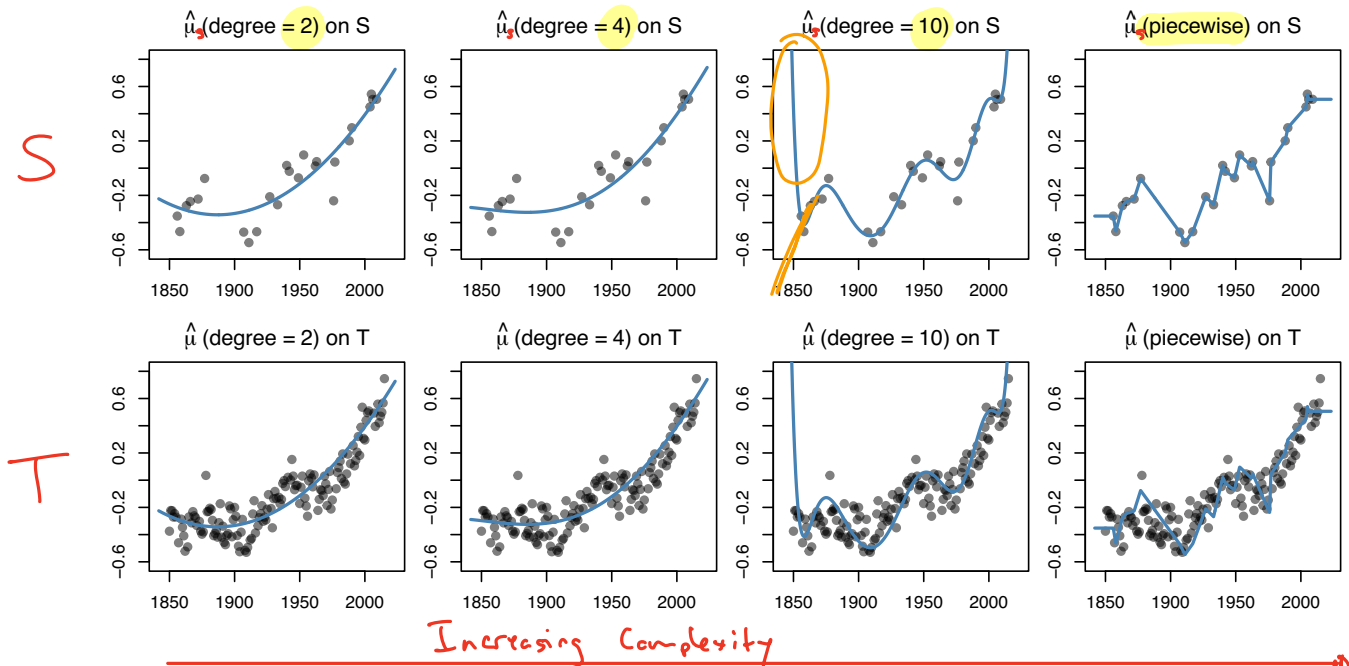
We find $APSE(S, \hat{\mu}_S) > APSE(T, \hat{\mu}_S)$ but this is unusual!

Typically, we'd expect it to be the other way around. This will become more apparent with more complicated models.

- Let's also consider some higher order polynomials

```
dset = c(2,4,10)
muhats = lapply(dset, getmuhat, sampleXY=sample.Data )
muhatFun = getmuFun(sample.Data, "x", "y" )

par(mfcol=c(2,4), mar=2.5*c(1,1,1,0.1))
for (i in 1:length(dset)) plotTemperatureSTfit(muhats[[i]], sample.temperature, dset[i])
plotTemperatureSTfit(muhatFun, sample.temperature)
```



- The APSE calculated over S and T for each model (and a function to do so) is shown below:

```
apseST <- function(y, x, sam, predfun) {
  apseS = apse(y[sam], x[sam], predfun)
  Tpop = !sam
  apseT = apse(y[Tpop], x[Tpop], predfun)

  c(apseS, apseT)
}

muhats[[4]] = muhatFun

temp.apse = sapply(muhats, apseST,
  y=temppop$y, x=temppop$x, sam=sample.temperature )

rownames(temp.apse) = c("APSE on S", "APSE on T")
colnames(temp.apse) = c(paste("deg=", dset), "piecewise linear ")
round(temp.apse,3)
```

"in-sample predictive accuracy"

	deg= 2	deg= 4	deg= 10	piecewise linear
## APSE on S	0.019	0.019	0.004	0.000
## APSE on T	0.015	0.015	0.024	0.016

"out-of-sample predictive accuracy"

- Note the difference in APSEs between $\mathcal{S}$ and $\mathcal{T}$.

Aside: Extrapolation Beyond the Sample

- In the previous plots the `getmuhat` function didn't work very sensibly for x values smaller or larger than what was observed in the sample
- We can modify the `getmuhat` function so that extrapolation beyond the minimum and maximum of the sample are set to constants.

```
getmuhat.ext <- function(sampleXY, complexity = 1) {
  formula <- paste0("y ~ ",
    if (complexity==0) "1"
    else {
      if (complexity < 3 ) {
        paste0("poly(x, ", complexity, ", raw = FALSE)")
        ## due to Numerical overflow
      } else {
        ## if complexity >= 20 use a spline.
        paste0("bs(x, ", complexity, ")")
      }
    }
  )

  fit <- lm(as.formula(formula), data = sampleXY)
  tx = sampleXY$x
  ty = fit$fitted.values

  range.X = range(tx)
  val.rY = c( mean(ty[tx == range.X[1]]),
    mean(ty[tx == range.X[2]]) )

  ## From this we construct the predictor function
  muhat <- function(x){
    if ("x" %in% names(x)) {
      ## x is a dataframe containing the variate named
      ## by xvarname
      newdata <- x
    } else {
      ## x is a vector of values that needs to be a data.frame
      { newdata <- data.frame(x = x) }
      ## The prediction
      ##
      suppressWarnings({
        ypred = predict(fit, newdata = newdata, silent = TRUE)
      })
    }
  }
}
```

Generally Speaking:

As model complexity increases

$APSE(\mathcal{S}, \hat{\mu}_{\mathcal{S}}) \downarrow$

$APSE(\mathcal{T}, \hat{\mu}_{\mathcal{S}}) \uparrow$

```

    #val = predict(fit, newdata = newdata)
    ypred[newdata$x < range.X[1]] = val.rY[1]
    ypred[newdata$x > range.X[2]] = val.rY[2]
    ypred
  }
  ## muhat is the function that we need to calculate values
  ## at any x, so we return this function from getmuhat
  muhat
}

```

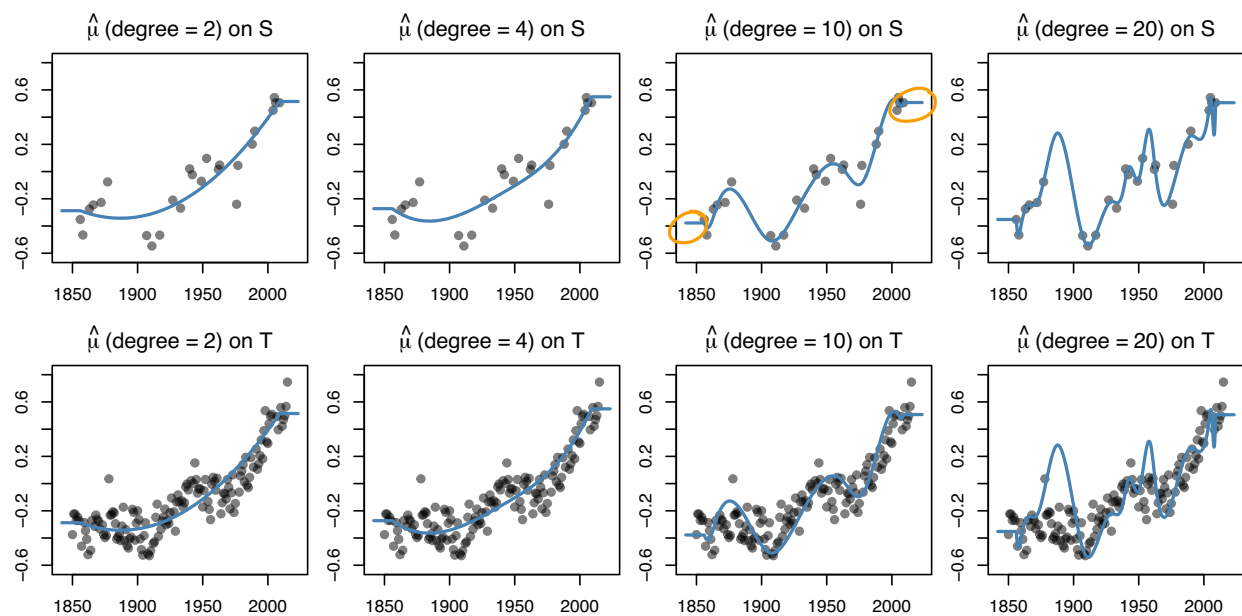
- Let's recreate the plots from above with this new version of muhat

```

dset = c(2, 4, 10, 20)
muhats = lapply(dset, getmuhat.ext, sampleXY=sample.Data )
muhatFun = getmuFun(sample.Data, "x", "y" )

par(mfcol=c(2,4), mar=2.5*c(1,1,1,0.1))
for (i in 1:length(dset)) plotTemperatureSTfit(muhats[[i]], sample.temperature, dset[i])

```



- Let us look at a larger collection of polynomials to pick an appropriate model.
- Comparing the APSE on the sample and complement set we have:

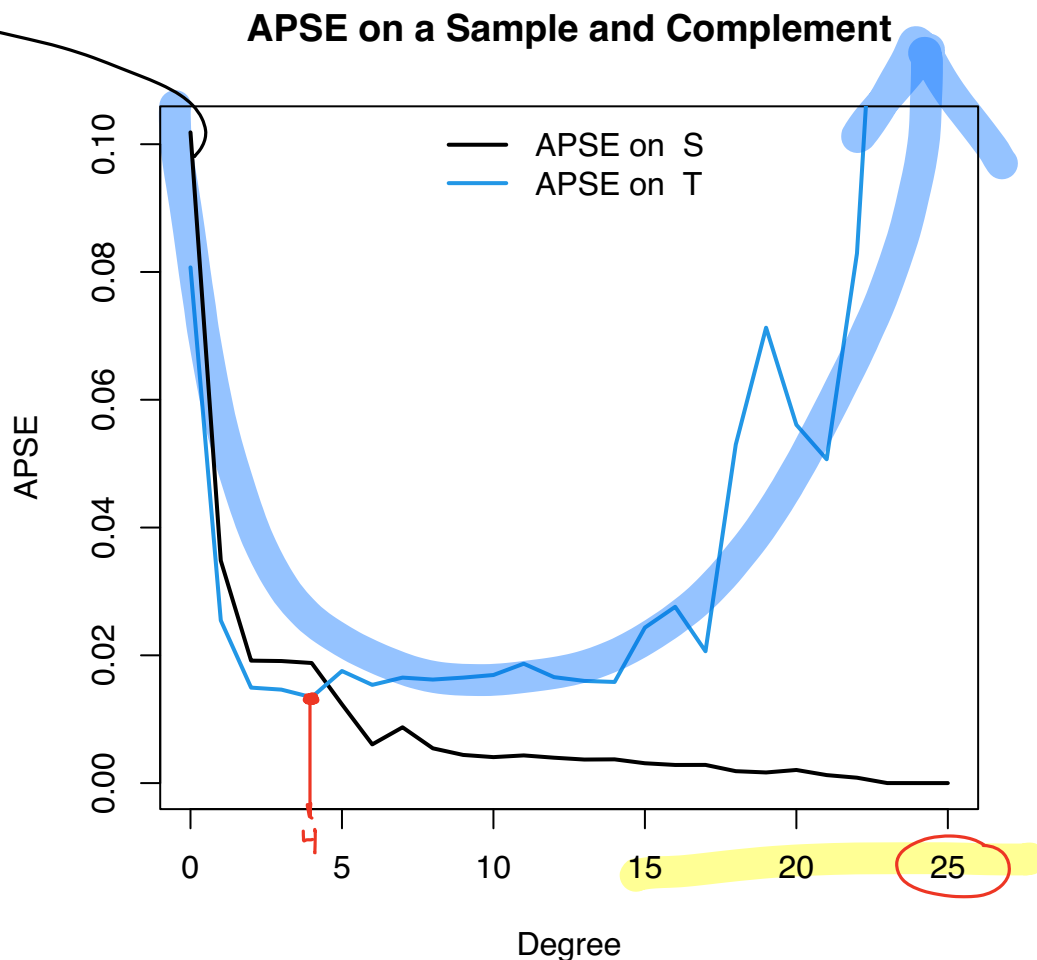
```
dset = 0:25
muhats = lapply(dset, getmuhat.ext, sampleXY=sample.Data )

temp.apse = sapply(muhats, apseST,
  y= temppop$y, x =temppop$x, sam=sample.temperature )

rownames(temp.apse) = c("APSE on S", "APSE on T")
colnames(temp.apse) = paste("deg=", dset)

plot( dset, temp.apse[1,], type='l', col=1, lwd=2,
  xlab="Degree", ylab="APSE",
  main="APSE on a Sample and Complement" )
lines(dset, temp.apse[2,], type='l', col=4, lwd=2 )
legend( "top", legend= paste("APSE on ", c("S","T")),
  col=c(1,4), lwd=c(2,2), bty='n')
```

Mimics what we saw previously on the green elbow plot



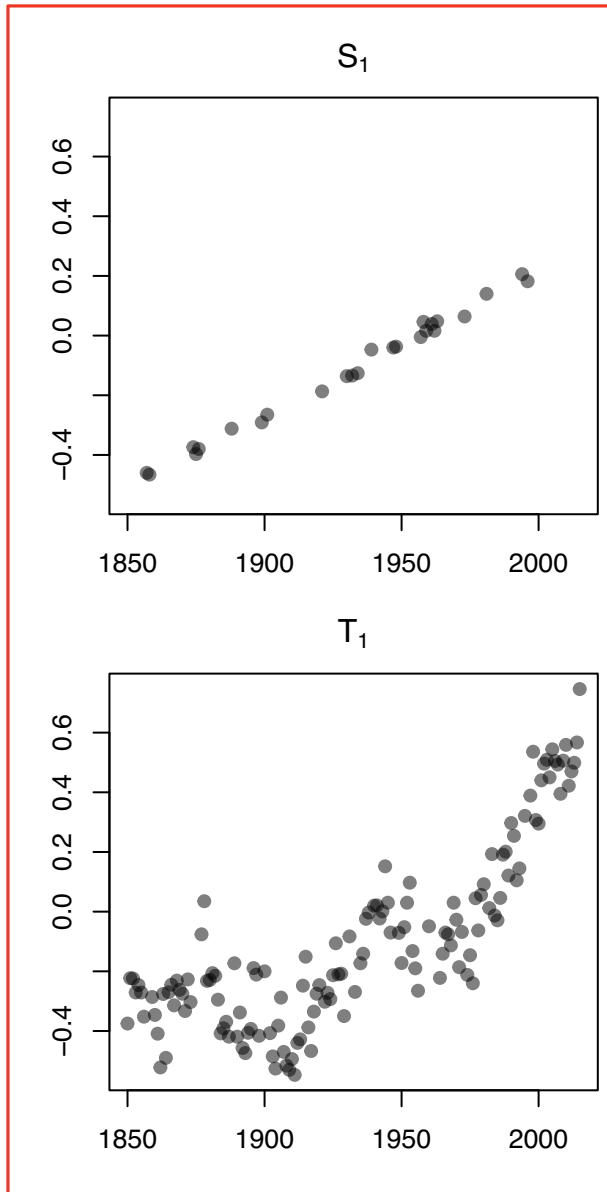
- At around degree equal to 4, increasing complexity does not improve prediction accuracy.
- ✱ Increasing complexity will continue to improve predictions on the sample but not on the rest of the population.
 - This effect is what we previously referred to as **overfitting**:
 - * the predictor function has been too closely tailored to the peculiarities of the sample.
 - * the complexity of the model has been increased so far that it has compromised the *out-of-sample* prediction performance

5.1.4 The Importance of a Good Sample

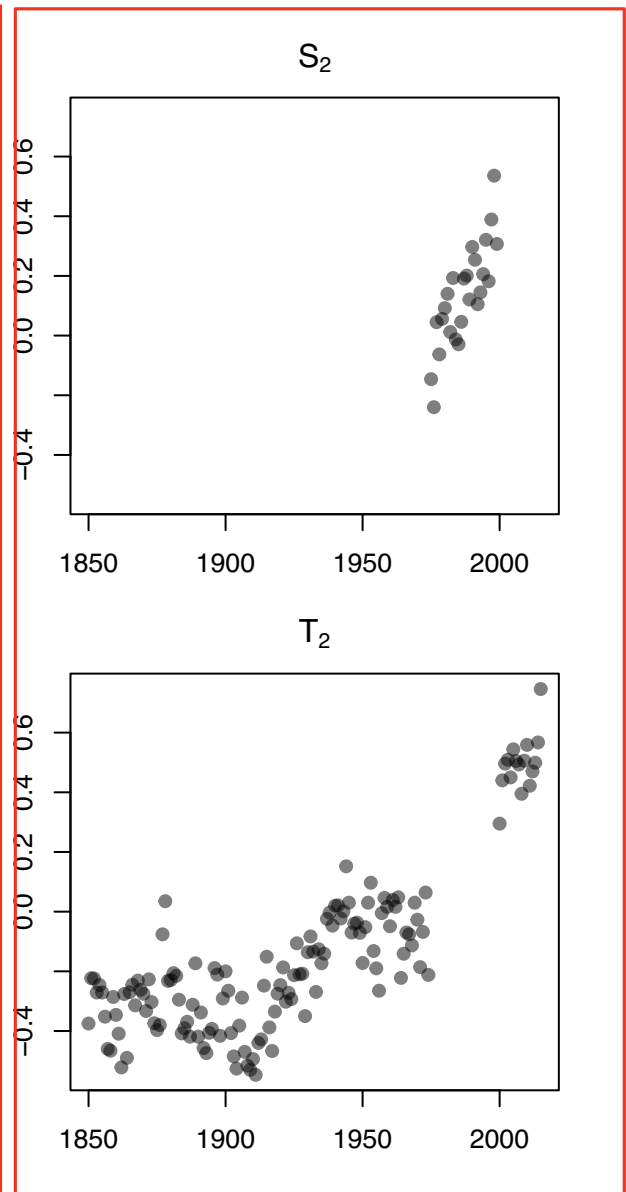
- ✱ Since $\hat{\mu}_{\mathcal{S}}(\mathbf{x})$ is based on a single sample $\mathcal{S}$ the quality of the predictor function depends crucially on the quality of the sample
 - ✱ if the sample is not a good/fair representation of the population, then any predictor function is bound to perform poorly
- ✱ It is important, then, to recognize that the performance (i.e., APSE) associated with $\hat{\mu}_{\mathcal{S}}(\mathbf{x})$ could vary quite a lot from one sample to another.

- In practice we tend to assume our sample is a good representation of the population
 - But in case that's not true it's important to choose a predictor function that performs well no matter which sample was used to estimate it.
 - Simpler is often better.
- Here are two partitions of the population $\mathcal{P}$:

Partition 1



Partition 2



- What do you observe?
- Are both samples *good* representations of the population?

No, S_1 is a better representation of P than S_2

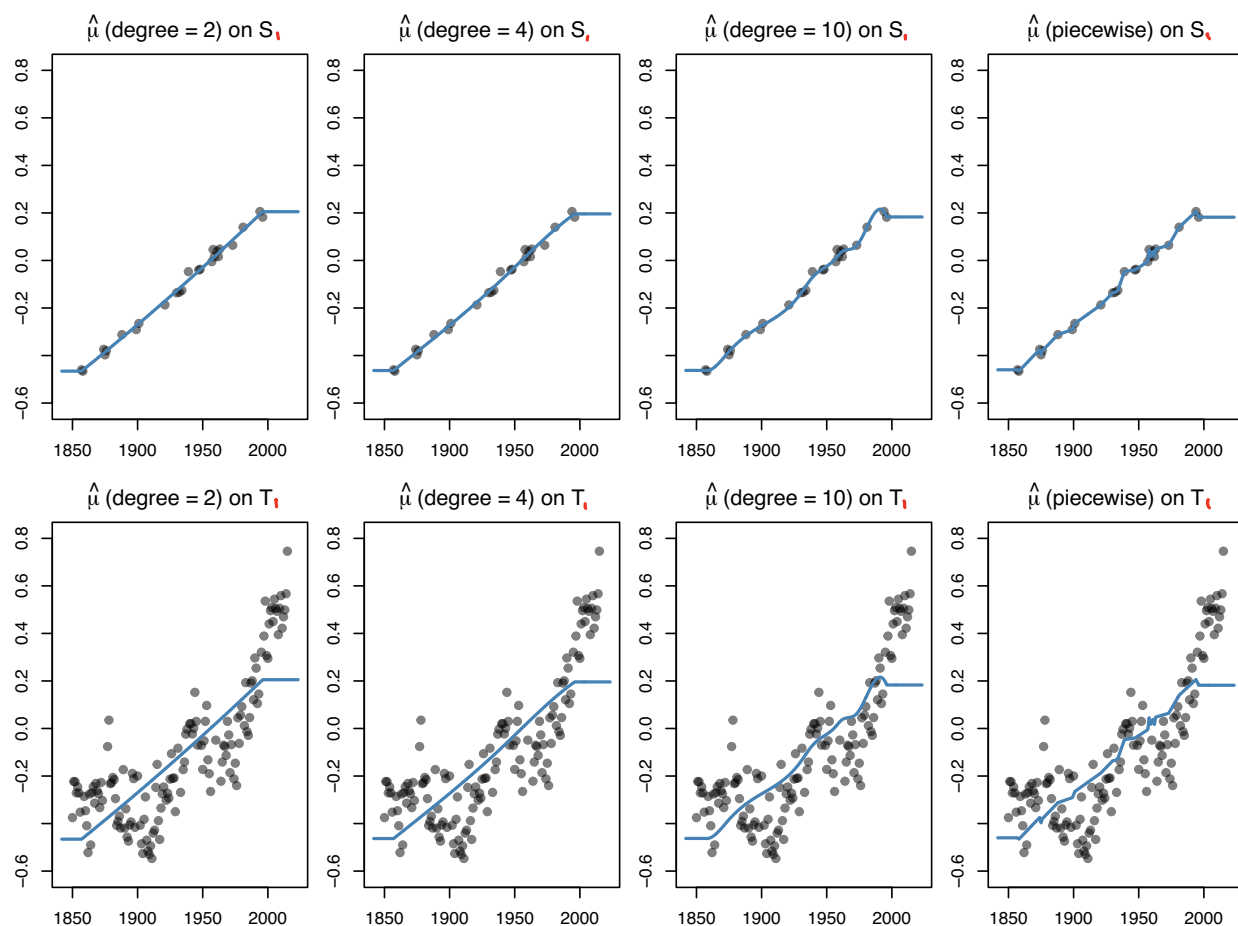
Example: Global Temperature Data (Sample 1)

- Here are some polynomials fit to Sample 1.

```
sam1.Data = getXYSample("YEAR", "ANNUAL", samp=sam1, pop=temperature)

dset = c(2,4,10)
muhats = lapply(dset, getmuhat.ext, sampleXY=sam1.Data )
muhatFun = getmuFun(sam1.Data, "x", "y" )

par(mfcol=c(2,4), mar=2.5*c(1,1,1,0.1))
for (i in 1:length(dset)) plotTemperatureSTfit(muhats[[i]], sam1, dset[i])
plotTemperatureSTfit(muhatFun, sam1)
```



- Which predictor performs better on the test set? *Degree 2 or 4 balances complexity and accuracy*
 - What would you have chosen had you not seen the test set?

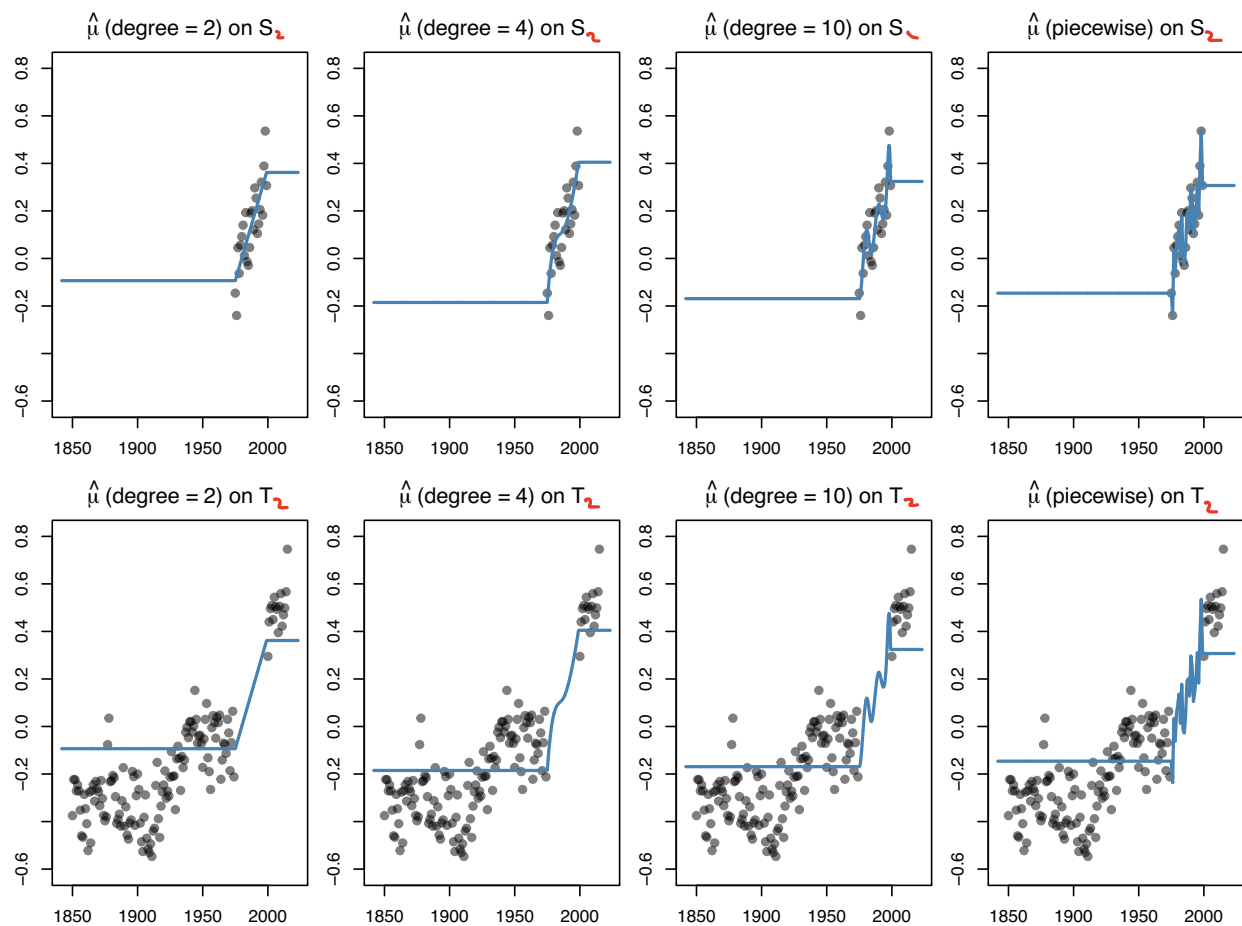
Example: Global Temperature Data (Sample 2)

- Here are some polynomials fit to Sample 2.

```
sam2.Data = getXYSample("YEAR", "ANNUAL", samp=sam2, pop=temperature)

dset = c(2,4,10)
muhats = lapply(dset, getmuhat.ext, sampleXY=sam2.Data )
muhatFun = getmuFun(sam2.Data, "x", "y" )

par(mfcol=c(2,4), mar=2.5*c(1,1,1,0.1))
for (i in 1:length(dset)) plotTemperatureSTfit(muhats[[i]], sam2, dset[i])
plotTemperatureSTfit(muhatFun, sam2)
```



- Which predictor performs better on the test set? *None of them*
 - What would you have chosen had you not seen the test set?